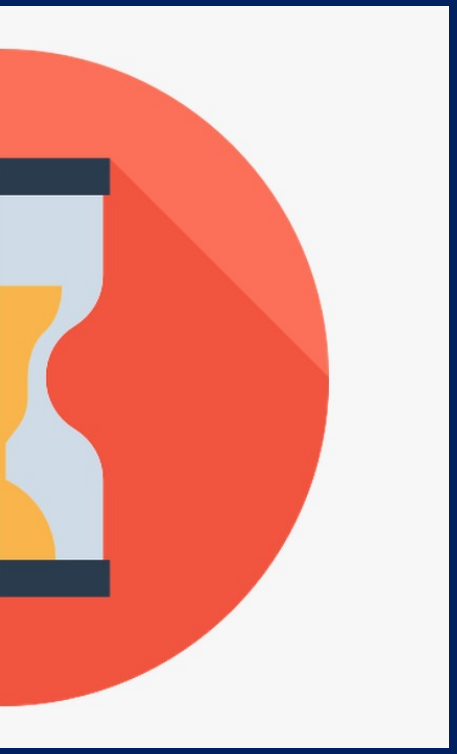


Finance-Growth Nexus



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Introduction

FINANCIAL SECTOR:

Financial sector is the set of institutions, instruments, markets as well as the legal and regulatory framework that permit transactions to be made by external credit.

IMPORTANCE:

- Financial sector development plays a huge role in economic development.
- It promotes economic growth through capital accumulation and technological progress by increasing the savings rate, mobilizing and pooling savings, producing information about investment, facilitating and encouraging the inflows of foreign capital, as well as optimizing the allocation of capital.





Research Question



- *Is it the change in Financial Development in the t'th period that leads to change in GDP in t+1?*

- *Or, is it the change in GDP in t'th period that leads to change in Financial-Development in t+1?*

Data Base

(DEVELOPED COUNTRIES)

- **SINGAPORE:**

Year	Depth	Efficiency	Stability	GDP (constant 2010 US$)
1996	93.74180643	2.43432	27.8414	111640540157.91
1997	96.47030493	2.08312	27.0812	120928833969.10
1998	107.9310737	1.62868	23.9321	118274049666.54
1999	102.7073041	1.93515	25.1285	125044059546.03
2000	96.05246317	2.18201	23.862	136347036849.47
2001	115.0239673	0.448513	12.1492	134889342486.59
2002	102.0242236	1.33686	6.67168	140169822003.39
2003	104.7828006	0.146868	15.1464	146527654292.44
2004	95.72896576	3.31811	24.9412	160915976665.76
2005	89.2233562	2.01508	26.9001	172757841656.22
2006	84.287056	1.86666	25.4967	188315010084.43
2007	85.3517596	1.89036	23.3988	205305009804.87
2008	97.86436233	2.22668	23.2418	209140543445.44
2009	96.86137655	1.93548	25.5952	209393611470.46
2010	94.8584027	1.77077	26.6419	239809387605.43
2011	104.7036738	1.60793	23.9712	254826893576.72
2012	112.9735432	1.65331	25.4517	266164946826.86
2013	124.0659516	1.59353	21.5825	278980908891.08
2014	128.1297633	1.51557	21.8843	289862763533.36
2015	122.4213538	1.41735	22.0707	298247041730.53
2016	124.0998189	1.38574	22.4277	307082095146.22
2017	122.7186498	1.87601	22.2063	318443461927.19

- **UNITED STATES (US):**

Year	Depth	Efficiency	Stability	GDP (constant 2010 US$)
1996	137.6657111	4.17162	21.9021	10602294994341.30
1997	146.5798201	4.18093	22.2583	11073801990034.60
1998	157.8046359	3.80936	22.7851	11570064188987.70
1999	171.6233134	4.31617	23.6337	12120016643945.60
2000	162.5981392	3.82926	23.5248	12620268393365.10
2001	170.851172	3.83036	24.9641	12746261681131.20
2002	162.2957763	3.94548	25.6634	12968262715349.60
2003	177.3723459	3.71041	25.6082	13339312044507.20
2004	184.8579609	3.50591	27.6959	13846057986118.70
2005	188.6700689	3.47201	27.623	14332499605579.60
2006	198.2978481	3.21106	27.7233	14741688498026.90
2007	206.6707388	3.13531	26.5905	15018267850123.00
2008	188.0974235	3.09929	22.9642	14997755929109.70
2009	194.2452267	3.4118	26.9327	14617299295858.10
2010	189.2309753	3.81113	28.9571	14992052727000.00
2011	180.5045594	3.64011	29.2992	15224554803720.80
2012	181.3929909	3.52514	29.3846	15567038144849.70
2013	193.5845545	3.42765	29.5357	15853795607833.00
2014	194.7161153	3.46118	29.6984	16242526401218.30
2015	189.072709	3.33533	29.9373	16710459044262.20
2016	191.9273137	3.3264	29.5835	16972347893377.00
2017	199.6396236	3.4207	29.7986	17348626599470.80

- **NORWAY:**

Year	Depth	Efficiency	Stability	GDP (constant 2010 US$)
1996	56.92234173	2.88676	8.47288	322361146092.53
1997	62.07790842	2.32715	7.9225	339396620604.66
1998	68.95823339	2.32553	8.36289	348303834512.28
1999	67.17787703	2.02442	8.62487	355314714530.52
2000	64.36820425	2.1866	8.37977	366702477304.83
2001	96.30705246	2.25201	8.50862	374310785078.08
2002	102.5308105	2.4884	7.44342	379724270302.26
2003	106.8351366	2.13249	7.21072	383180162395.03
2004	105.3196595	1.37269	8.33642	398390171552.69
2005	106.9547904	1.68918	8.27171	408849736694.05
2006	97.52964039	2.03731	6.98191	418661275147.65
2007	107.5196166	1.48522	6.95378	431197048129.25
2008	119.9358672	1.28388	5.89478	433251184624.93
2009	128.1241237	1.40673	6.33215	425768848675.37
2010	128.3207003	1.37459	7.36659	428757038466.84
2011	125.7499688	1.5135	7.40629	432965158016.92
2012	123.4714426	1.61671	7.81932	444668526003.85
2013	124.9830035	1.3582	8.32743	449266722786.28
2014	129.1008955	1.24328	8.57747	458115229920.28
2015	138.6784963	1.25708	9.99681	467126944887.83
2016	145.8085927	1.44316	10.6857	472132485600.73
2017	146.4868716	2.01086	11.1452	483101343264.08

(DEVELOPING COUNTRIES)

- **INDIA:**

Year	Depth	Efficiency	Stability	GDP (constant 2010 US$)
1996	23.40244204	3.95959	10.5823	699374141678.86
1997	23.55532535	3.92807	13.1166	727697541481.73
1998	23.67774686	3.55741	13.6532	772701383365.47
1999	25.42286221	3.39156	12.2518	841052658954.87
2000	28.33955116	3.26656	12.9598	873357417209.47
2001	28.61941543	3.36064	11.7536	915487884378.98
2002	32.30648907	3.04702	11.8042	950312817570.94
2003	31.62626562	3.34201	13.1828	1025011030242.90
2004	36.19180385	3.71943	14.4729	1106222004443.92
2005	40.06798048	3.52568	14.643	1193872737485.77
2006	43.62775243	3.37657	15.5567	1290107626116.85
2007	45.62776475	3.33443	14.9892	1388940385493.75
2008	49.55936669	3.14115	17.4858	1431812781420.86
2009	48.12444791	2.73093	16.7469	1544380310593.35
2010	50.55537498	3.08479	17.2801	1675615335600.56
2011	51.2892197	3.34939	16.9837	1763440111904.05
2012	51.88850765	3.06815	17.1177	1859659734290.56
2013	52.38570952	3.01061	17.2634	1978419583617.95
2014	51.88218736	2.85019	16.538	2125024977747.68
2015	51.86752408	2.89992	16.4565	2294947367978.21
2016	49.19481968	2.84155	17.5591	2482433701491.81
2017	48.77985713	3.01258	15.9562	2660371790259.41

- **BRAZIL:**

Year	Depth	Efficiency	Stability	GDP (constant 2010 US$)
1996	40.77849368	7.46815	11.8877	1414207205945.88
1997	40.85223717	6.8115	13.4714	1462217362494.31
1998	29.53226057	6.89881	15.6367	1467161088733.51
1999	29.8269102	6.60212	17.9604	1474026486641.66
2000	31.14098578	6.39964	15.7966	1538706023647.11
2001	29.00403802	6.08824	16.8682	1560092443326.51
2002	29.64514194	6.42769	17.9645	1607729271032.29
2003	27.68566994	8.89114	18.5254	1626070712767.17
2004	29.37277026	7.47037	17.7455	1719731810823.10
2005	31.83978415	7.99984	17.3913	1774799882787.63
2006	35.42516667	8.29784	18.5868	1845117253979.08
2007	40.69093794	7.84769	17.4945	1957113483856.30
2008	45.78225599	3.5381	17.508	2056812669849.93
2009	47.49428985	7.42687	16.9358	2054224952705.78
2010	52.76465763	6.3198	15.1804	2208871646202.82
2011	58.07850438	5.37245	14.5038	2296661602087.89
2012	62.5192932	5.49566	13.0293	2340783923756.88
2013	64.23058587	3.83322	12.5102	2411121328917.38
2014	66.02849897	3.64969	12.847	2423272459124.27
2015	66.82970064	3.18268	15.4336	2337346329824.16
2016	62.19062192	8.33638	15.4676	2260088660024.86
2017	59.77144021	6.26847	15.2382	2284132218956.82

- **MALAYSIA:**

Year	Depth	Efficiency	Stability	GDP (constant 2010 US$)
1996	141.6337391	3.60478	11.8329	141478022476.79
1997	158.3850476	3.27034	12.2812	151838092825.43
1998	158.5053788	2.46287	7.68778	140663697153.15
1999	149.1529811	1.91295	12.3823	149297089131.04
2000	134.9998539	2.27636	15.1763	162523121449.19
2001	129.1013756	2.73102	13.21	163364463537.30
2002	121.8273449	2.76622	13.6085	172171422650.65
2003	118.9741489	2.61957	13.396	182137564217.19
2004	111.9376051	3.61407	15.2595	194492752475.86
2005	106.5244319	2.51358	14.4259	204863376672.57
2006	103.6638304	1.5231	12.3796	216304682965.45
2007	101.5800752	1.90485	12.4405	229929251901.53
2008	96.74825351	2.29031	13.2883	241038904256.31
2009	111.60689	2.58771	13.2761	237390711216.67
2010	107.1227526	2.83612	13.8204	255016609232.87
2011	108.4259367	4.93651	18.9284	268516966253.76
2012	114.1245298	2.87679	17.5624	283214119400.21
2013	119.8996939	2.52338	17.3823	296507404302.88
2014	120.578651	2.44906	18.0114	314317779640.49
2015	123.1036492	1.80373	15.0995	330321318804.14
2016	121.9797907	1.92707	16.4246	345019810731.17
2017	117.2241027	2.25661	23.3917	364830260267.54

Period Of Study

We have taken 1996-2017 as our period of study. Factors that lead to us choosing these period of study is because of the important economic phenomenon of these selected countries.

- The impact of the Asian Financial Crisis that hit Malaysia's economy(1996-2003) and it's recovery from it to emerge as the world's largest Islamic banking and financial centre.
- The effect of the Economic Liberalisation in India (1991) and the start of the economic slowdown that emerged after the end of the 20 years long boom period in 2012.
- The introduction of the historic inflation stabilization program Plano Real(Real Plan) and its impact (ended in 2010) and the economic contraction that started thereafter.
- The 1990's economic boom in the United States(an economic expansion that ended in 2001);the start of the early 2000's recession during the Dot com Bubble crash(2000-02)
- The rapid transformation of Singapore as an industrial hub for international trade despite the obstacles in the form of Severe Acute Respiratory Syndrome(SARS) outbreak, small labour force etc.

SOURCE OF OUR DATA:- WORLD BANK'S WORLD DEVELOPMENT INDICATOR(WDI) DATABASE

Indicators

FINANCIAL INDICATORS:

- **Depth:- Private Sector Credit to GDP**

The ratio of private credit by deposit money banks to GDP(%).

Depth measures the size of financial intermediaries.

- **Efficiency:- Net interest margin(%)**

Efficiency ratios are metrics that are used in analysing a country's ability to effectively use it's resources to produce income.

- **Stability:- Bank Z-score**

Bank non performing loans to gross loans(%). Understanding how many defaulting loans a bank is holding allows us to explain the role that *Depth* have on the overall health of the financial intermediaries of the country.

GROWTH INDICATOR:

- **Real GDP:-**

Real Gross Domestic Product(real GDP) is a macroeconomic measure of the value of economic output adjusted for price changes (that is inflation or deflation). This adjustment transforms the money-value measure, nominal GDP, into an index for quantity of total output.

Methodology

We have used charts and correlation coefficients of the financial indicators and real GDP of both developed and developing countries to compare the results of our study. Here, we have taken:

t'th period from 1996 to 2016 and

(t+1)th period from 1997 to 2017.

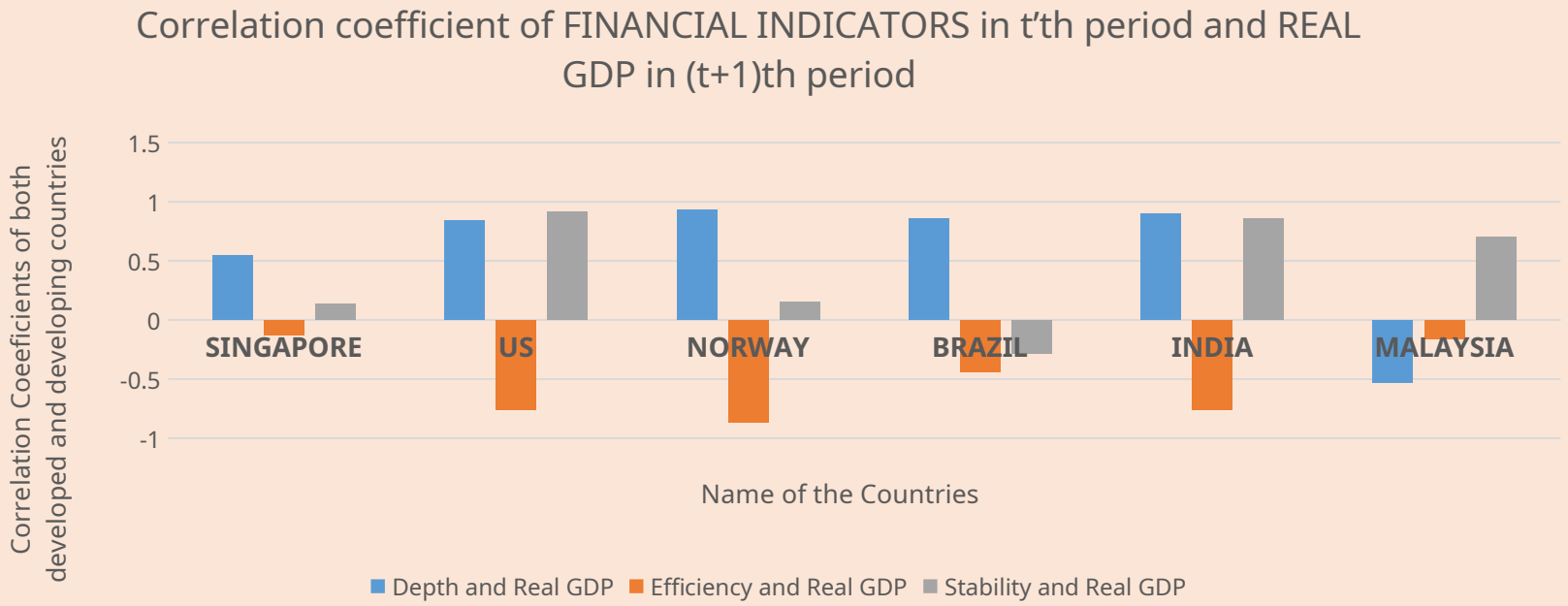
Comparison of the correlation coefficients

- Correlation coefficient when financial indicator is in t'th period whereas Real GDP is in (t+1)th period :-

Name of the countries	Depth and Real GDP	Efficiency and Real GDP	Stability and Real GDP
DEVELOPED:			
SINGAPORE	0.548038	-0.12534	0.135994
UNITED STATES(US)	0.842517	-0.75569	0.917732
NORWAY	0.934096	-0.86122	0.148936
DEVELOPING:			
BRAZIL	0.86104834	-0.438859601	-0.282959736
INDIA	0.89838372	-0.758182401	0.855212955
MALAYSIA	-0.527405561	-0.157528594	0.704272172

TABLE 1

- Graphical Representation:-

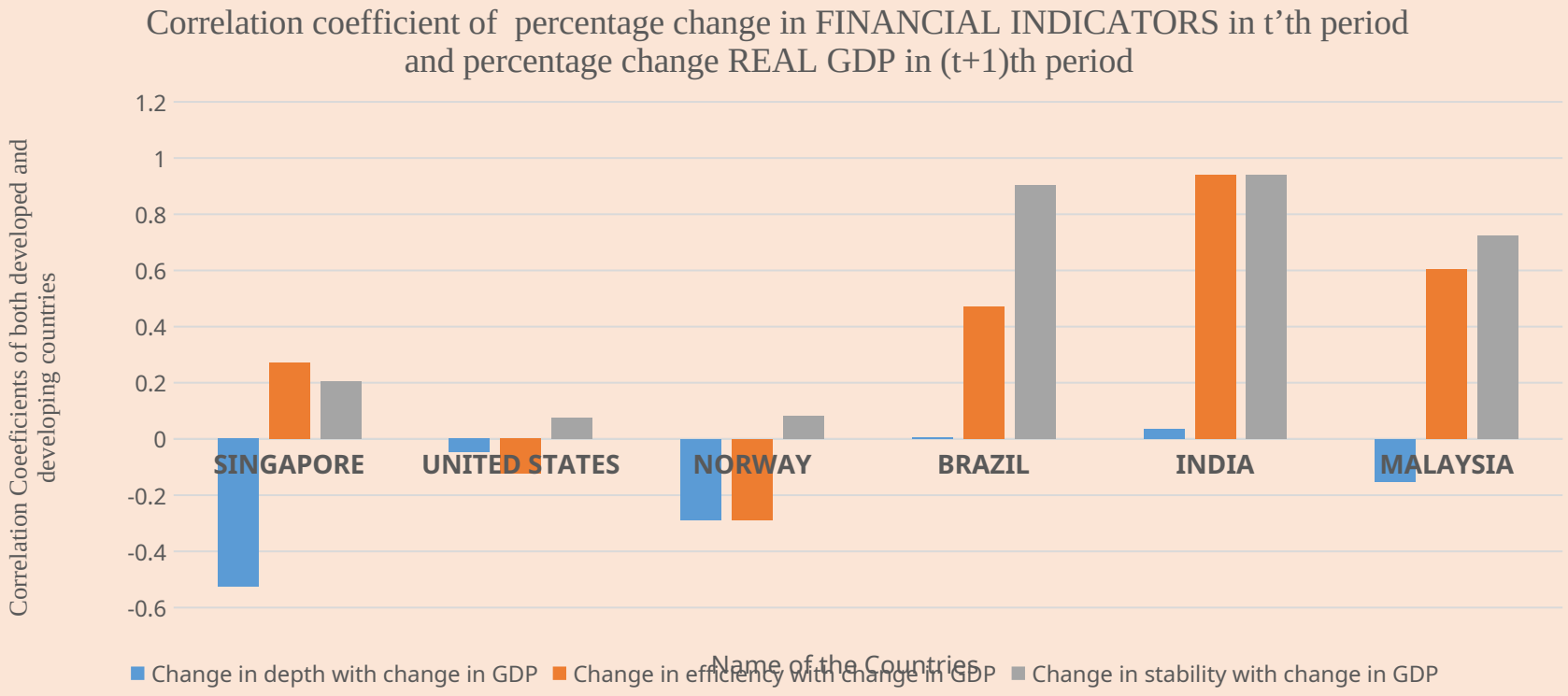


- Correlation coefficient of percentage change in financial indicator in t’th period and percentage change in Real GDP in (t+1)th period :-

COUNTRIES	Change in depth with change in GDP	Change in efficiency with change in GDP	Change in stability with change in GDP
DEVELOPED:			
SINGAPORE	-0.52596	0.272047	0.205307
UNITED STATES(US)	-0.046969854	-0.123302631	0.075532874
NORWAY	-0.287644332	-0.28764	0.082488
DEVELOPING:			
BRAZIL	0.00369	0.471367	0.904232
INDIA	0.036358	0.939473	0.940577
MALAYSIA	-0.15068	0.603653	0.724969

TABLE 2

- Graphical Representation:-

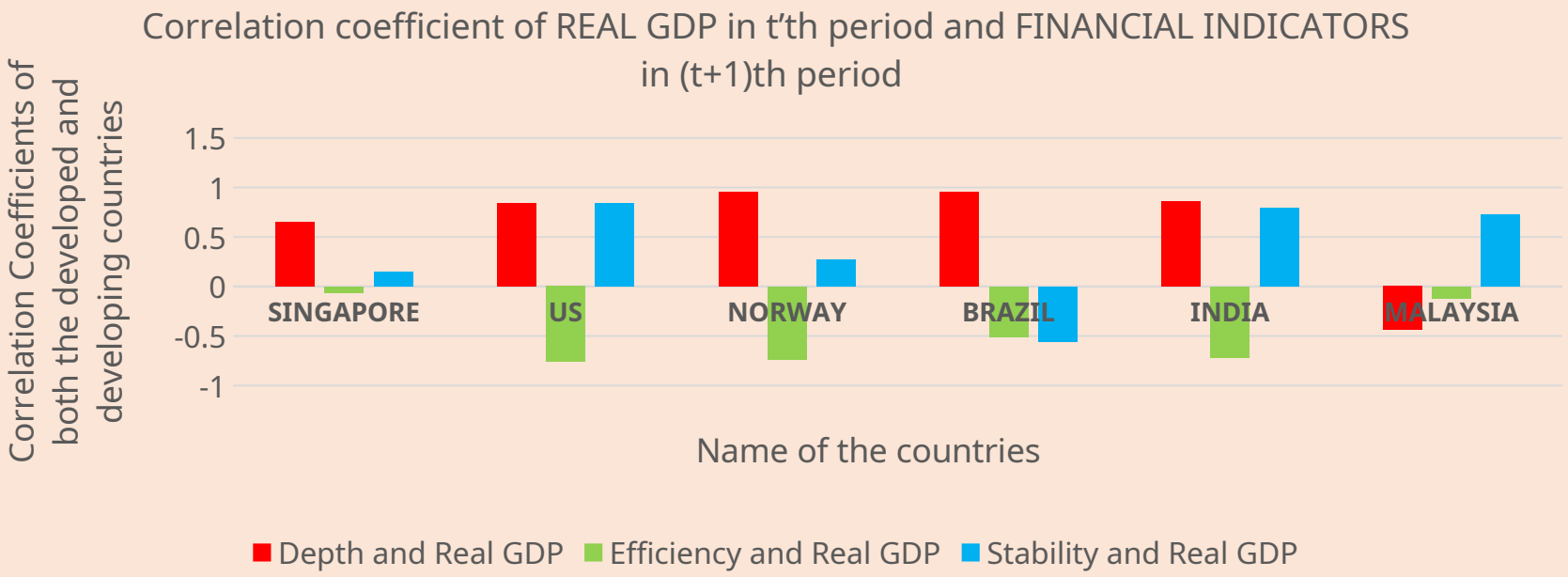


- Correlation coefficient of Real GDP in t'th period and financial indicators in (t+1)th period :-

Name of the countries	Depth and Real GDP	Efficiency and Real GDP	Stability and Real GDP
DEVELOPED:			
SINGAPORE	0.648723	-0.06237	0.147418
UNITED STATUS(US)	0.839947	-0.76217	0.843443
NORWAY	0.953543	-0.73966	0.266665
DEVELOPING:			
BRAZIL	0.953441037	-0.510757536	-0.557241607
INDIA	0.858097163	-0.716857556	0.793352526
MALAYSIA	-0.438683825	-0.118514714	0.728969535

TABLE 3

- Graphical Representation:-

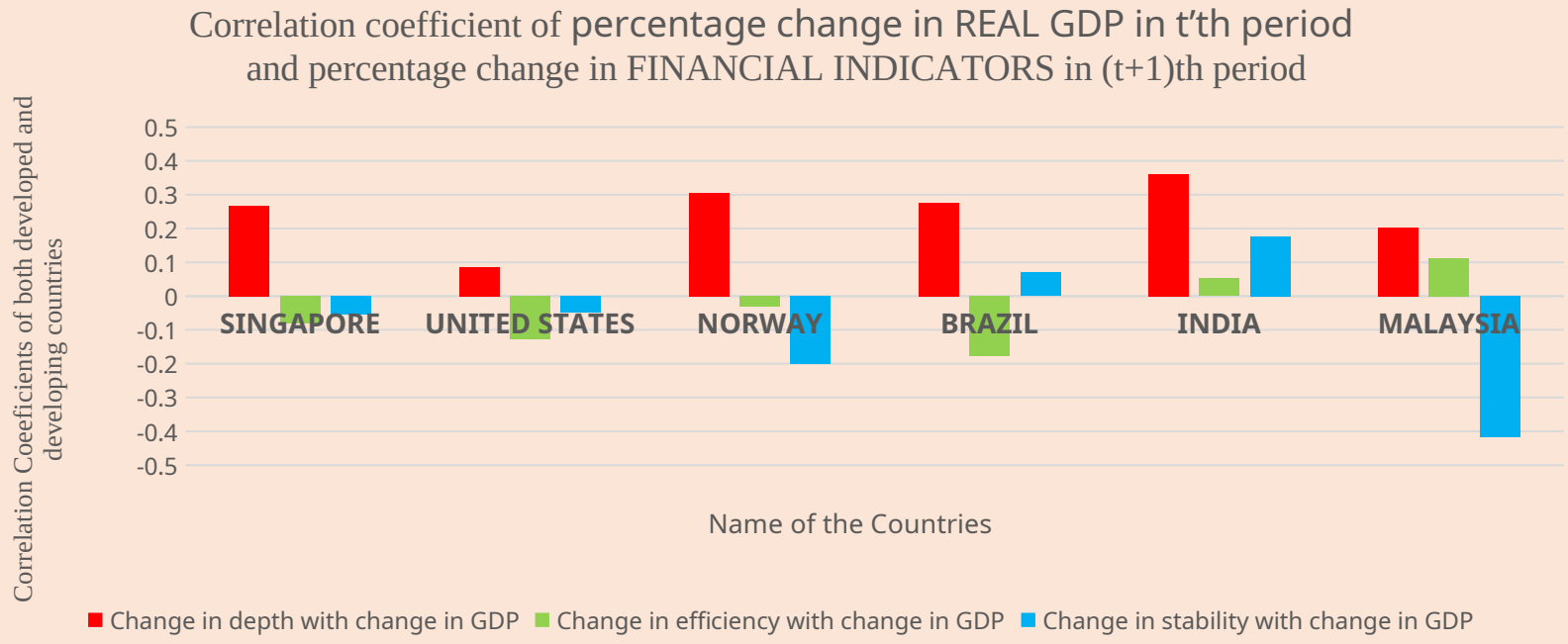


- Correlation coefficient of percentage change in Real GDP in t'th period and percentage change in financial indicator in (t+1)th period :-

COUNTRIES	Change in depth with change in GDP	Change in efficiency with change in GDP	Change in stability with change in GDP
DEVELOPED:			
SINGAPORE	0.2665	-0.08133	-0.05375
UNITED STATES(US)	0.085087	-0.12903	-0.04925
NORWAY	0.30643	-0.03175	-0.20019
DEVELOPING:			
BRAZIL	0.27507	-0.17696	0.06961
INDIA	0.36112	0.05189	0.17653
MALAYSIA	0.20207	0.1119	-0.4162

TABLE 4

- Graphical Representation:-



Analysis

- The debate on the role of finance in economic development has been an ongoing one, especially in developing countries.
- Here we are comparing and analyzing the correlation between financial indicators and real GDP with respect to developed and developing countries.

➢ **Financial Depth(Private Sector Credit to GDP)**

Lagged financial indicator (t^{th} period financial indicator vs. $(t+1)^{th}$ period GDP) (Refer TABLE 1)

- In the $(t+1)^{th}$ year, the correlation coefficient between a lagged private sector credit to GDP and a current real GDP is positive for all the developed countries Singapore, the US and Norway with all values way above 0 and even 0.5, which explains that in such countries, real GDP directly depends on previous year's financial depth indicator. In fact, it is very close to 1 (0.93) in case of Norway, which shows its heavy dependence.
- The same for developing countries cannot be uniformly said for all developing countries. While it is heavily positive for Brazil and India which depicts a direct relationship, it is negative for Malaysia. But overall it can be concluded that there is a positive relationship between lagged private sector credit to GDP and current real GDP.

Lagged real GDP (t^{th} period GDP vs. $(t+1)^{th}$ period financial indicator) (Refer TABLE 3)

- In the $(t+1)^{th}$ year, the correlation coefficient between a lagged real GDP and current private sector credit to GDP is positive for all developed countries , and all the values are significantly high, which depicts a heavy dependence.
- However, there is no uniform positive relationship between current private sector credit to GDP and previous year's real GDP for developing countries. However if we ignore the low negative value of the said correlation coefficient in case of Malaysia, we can fairly conclude that current private sector credit to GDP depends directly and heavily on previous year's real GDP.

Change in Depth vs. Change in Real GDP(Refer TABLE 2 & 4)

- In case of developed countries the correlation coefficient between relative change of depth and relative change of real GDP is negative , which implies an inverse relationship between the two.
- In case of developing countries the correlation coefficient between relative change of depth and relative change of real GDP is mostly positive (except for a negative value for Malaysia) and hence one can say that overall it is a direct relationship with values very close to 0 but positive.

➢ Financial Efficiency(Net Interest Margin (%))

Lagged financial indicator (t^{th} period financial indicator vs. $(t+1)^{th}$ period GDP) (Refer TABLE 1)

- In the (t+1)th year, the correlation coefficient between a lagged net interest margin % and the current real GDP is negative for all the three developed countries, and it is quite large for most of them. This implies that lagged efficiency indicator and current real GDP are inversely related.
- Again, the same thing i.e. an inverse relationship can be concluded from the negative values of the correlation coefficients between lagged efficiency indicator and current real GDP for developing countries.

Lagged real GDP (t^{th} period GDP vs. $(t+1)^{th}$ period financial indicator) (Refer TABLE 3)

- In the $(t+1)^{th}$ year, the correlation coefficient between a lagged real GDP and the current net interest margin % is negative for all the three developed countries, which shows that there is a negative relationship between lagged efficiency indicator and current real GDP.
- Similarly, there are negative values of the same correlation coefficient for all the three developing countries, from which a negative relationship is concluded to exist between lagged real GDP and current financial efficiency indicator.

Change in Efficiency vs. Change in Real GDP(Refer TABLE 2 & 4)

- In case of developed countries the correlation coefficient between relative change of efficiency and relative change of real GDP is negative mostly . If we ignore the positive correlation coefficient for Singapore, then one can conclude that an inverse relationship exists.
- In case of developing countries the correlation coefficient between relative change of efficiency and relative change of real GDP is positive, which implies a direct relationship between the two.

➢ Financial Stability (Bank Z-Score)

Lagged financial indicator (t^{th} period financial indicator vs. $(t+1)^{th}$ period GDP) (Refer TABLE 1)

- In the $(t+1)^{th}$ year, the correlation coefficient between a lagged bank Z-score and the current real GDP is positive, with most values closer to 0 than 1.This implies a light dependence of GDP on lagged stability indicator.
- However a uniform positive relationship between lagged stability indicator and the current real GDP cannot be claimed for all developing countries. However if we ignore the negative value of the said correlation coefficient for Brazil, then a direct relationship can be concluded.

Lagged real GDP (t^{th} period GDP vs. $(t+1)^{th}$ period financial indicator) (Refer TABLE 3)

- In the $(t+1)^{th}$ year, the correlation coefficient between a lagged real GDP and the current bank Z-score is positive for all three chosen developed countries, which implies a direct relationship between the said variables.
- The same for developing countries cannot be uniformly said for all developing countries. While it is heavily positive for Malaysia and India which depicts a direct relationship, it is negative for Brazil. But overall it can be concluded that there is a positive relationship between lagged bank Z-score and current real GDP.

Change in Stability vs. Change in Real GDP(Refer TABLE 1)

- In case of developed countries the correlation coefficient between relative change of stability and relative change of real GDP is positive, hence one can conclude that a direct relationship exists.
- In case of developing countries the correlation coefficient between relative change of stability and relative change of real GDP is positive, which implies a direct relationship between the two.

Conclusion

- ✓ Six countries were chosen i.e. three developed (Singapore, United States(US) and Norway) and three developing (Brazil, India and Malaysia) to evaluate and analyse the relationship between real GDP and financial indicators i.e. depth (private sector credit to GDP), efficiency (net interest margin %) and stability (bank Z-score).
- ✓ The research question was to determine whether the (t+1)-th period real GDP depends on t-th period financial indicator or whether the (t+1)-th period financial indicator depends on t-th period real GDP.
- ✓ The methodology that has been used are correlation coefficients between real GDP and financial indicator.
- ✓ By comparing the correlation coefficients for lagged financial indicator-current GDP model and for lagged GDP- current financial indicator model, the data findings showed that correlation coefficients for the latter model were higher than those of the former model.
- ✓ Now since a higher correlation coefficient indicates that there is a greater goodness of fit for the model, hence we conclude that the (t+1)-th period financial indicator depends on the t-th period real GDP, ie. real GDP is the independent variable, and the three given financial indicators depend on it.

THANK YOU 

AREAS OF CONTRIBUTION	NAME OF THE GROUP MEMBERS
Data collection & data selection	Shreya Ghosh, Vasundhara Chatterjee, Pronay Ghosh
Charts	Avike Paul, Moulina Bandyopadhyay, Vasundhara Chatterjee, Shreya Ghosh
Tables	Avike Paul, Vasundhara Chatterjee
Analysis & Conclusion	Abhishikta Kundu
Powerpoint Presentation	Moulina Bandyopadhyay, Avike Paul